

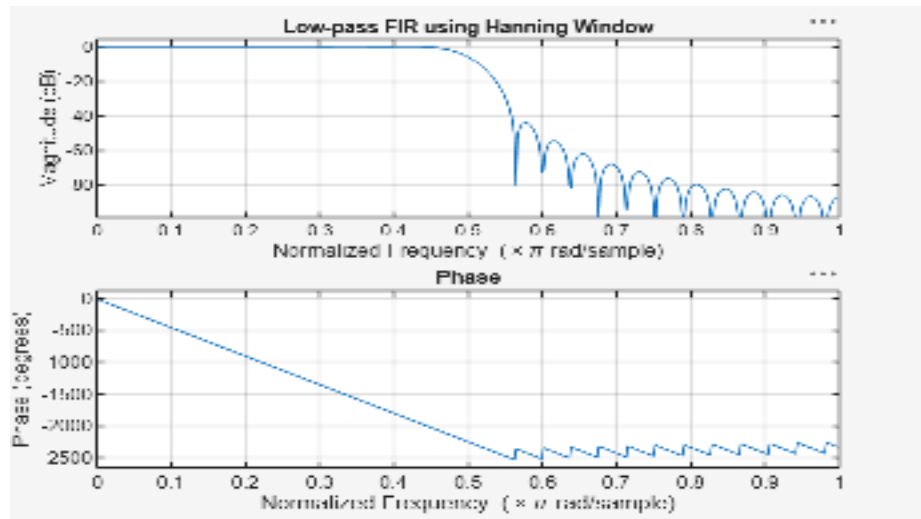
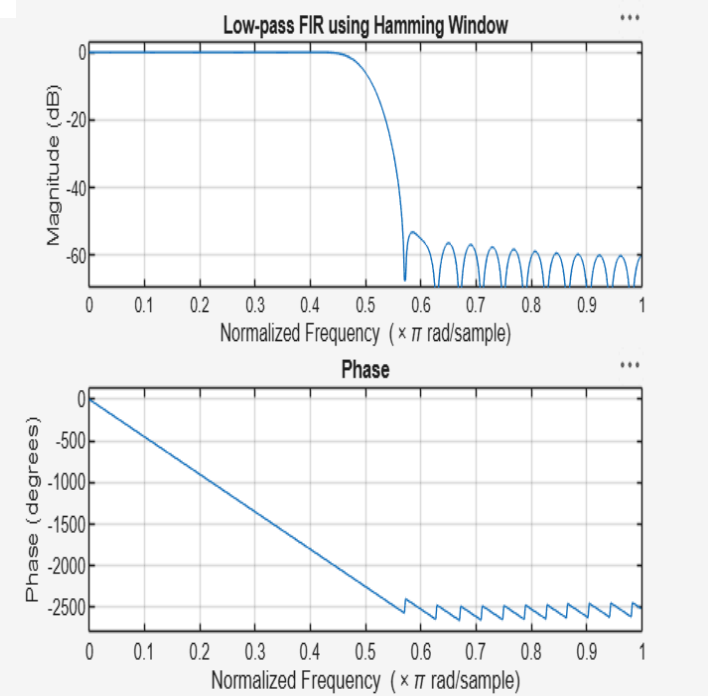
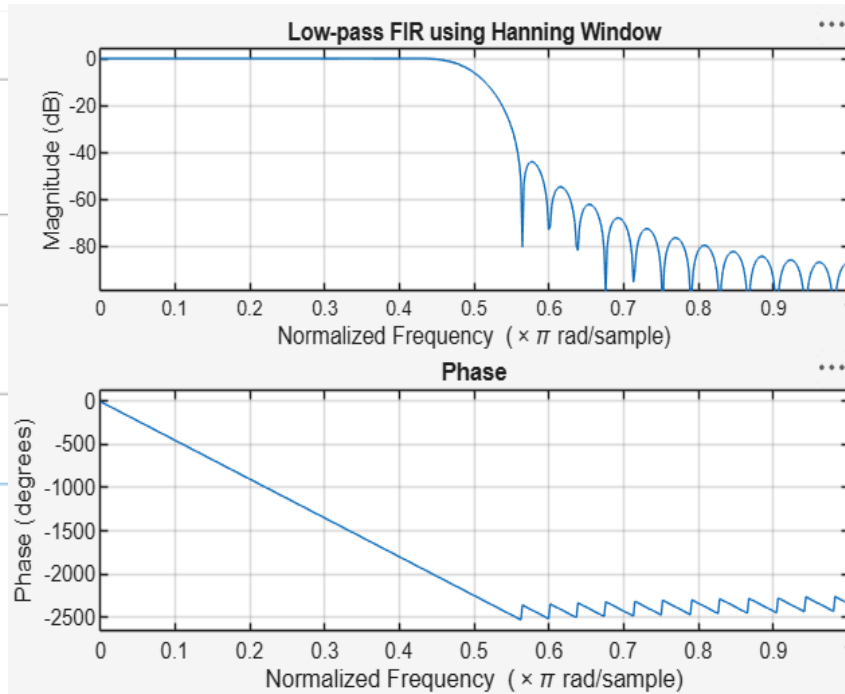
EXP 4 .1

```

clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm');
%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann');
%% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

```



Exp 4.2

```
clc;
clear;
close all;

% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)

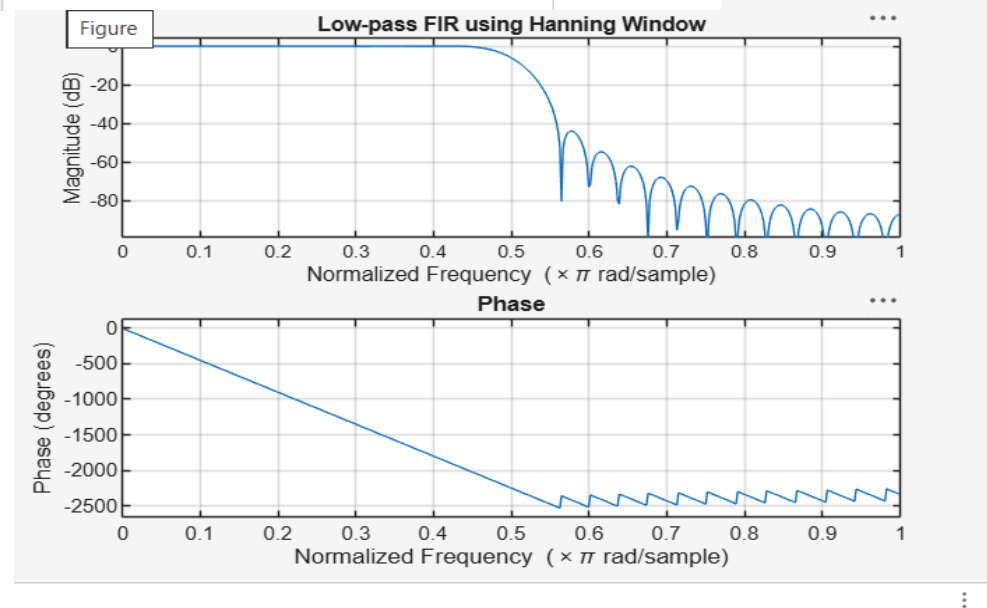
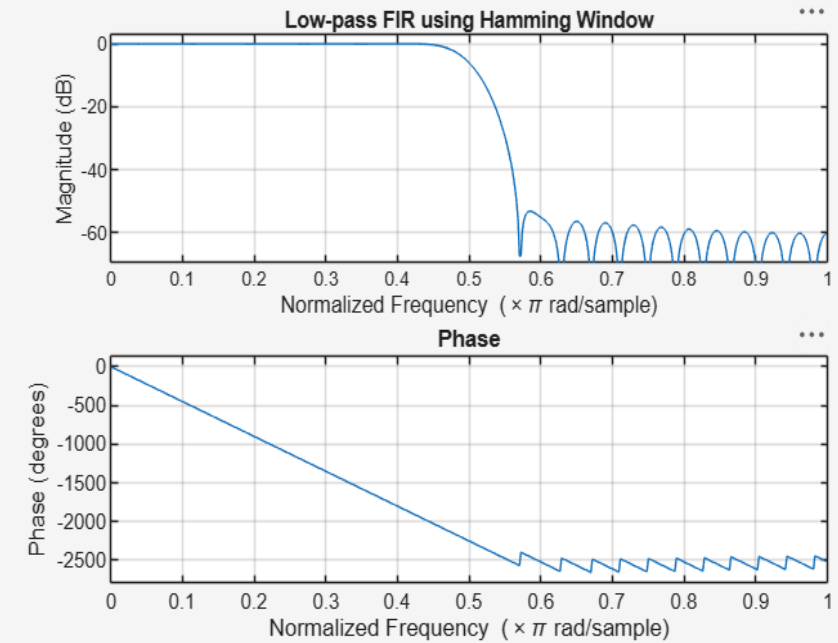
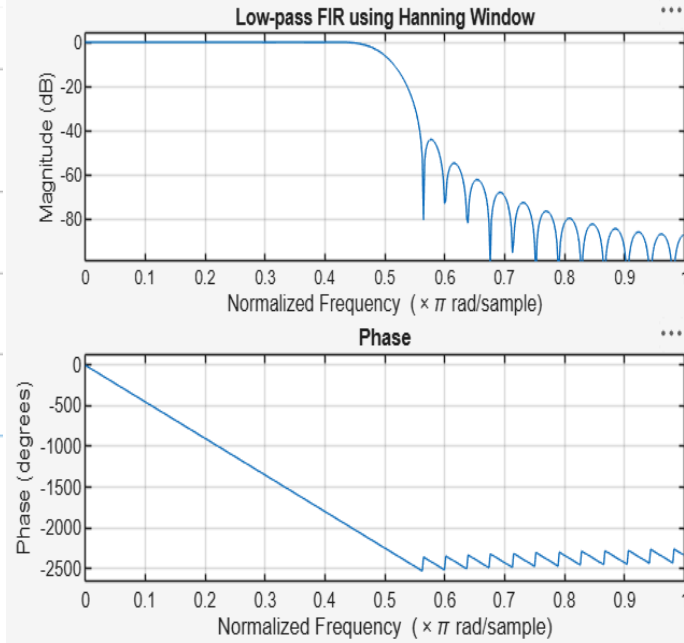
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);

% --- Plot Frequency Responses ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```



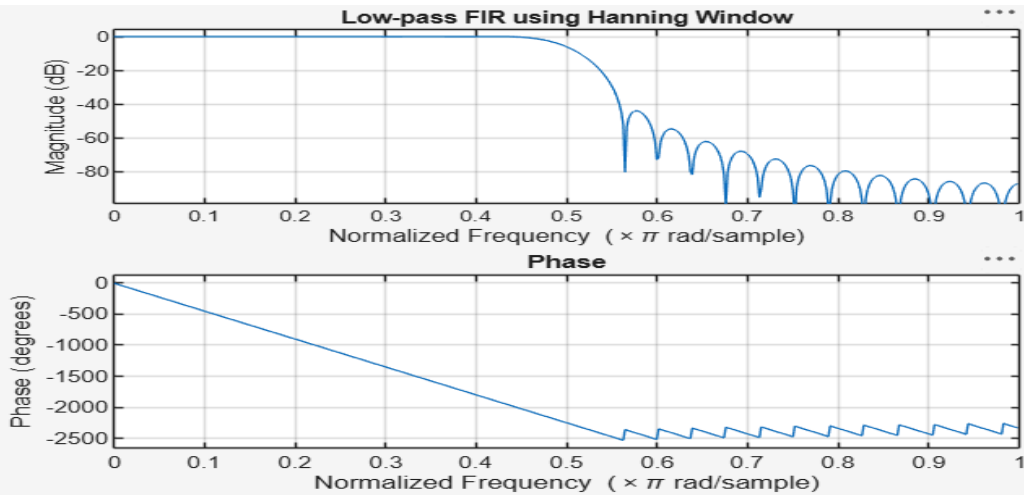
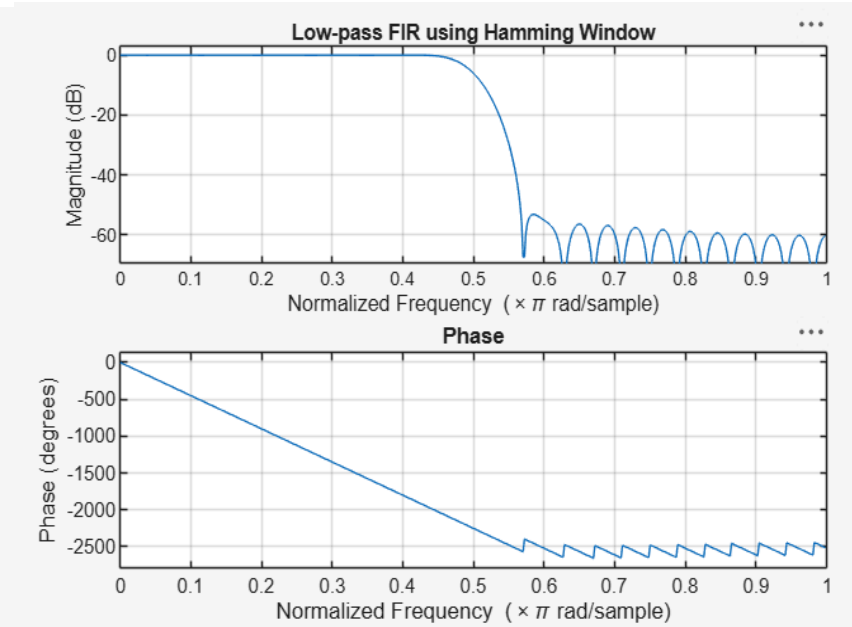
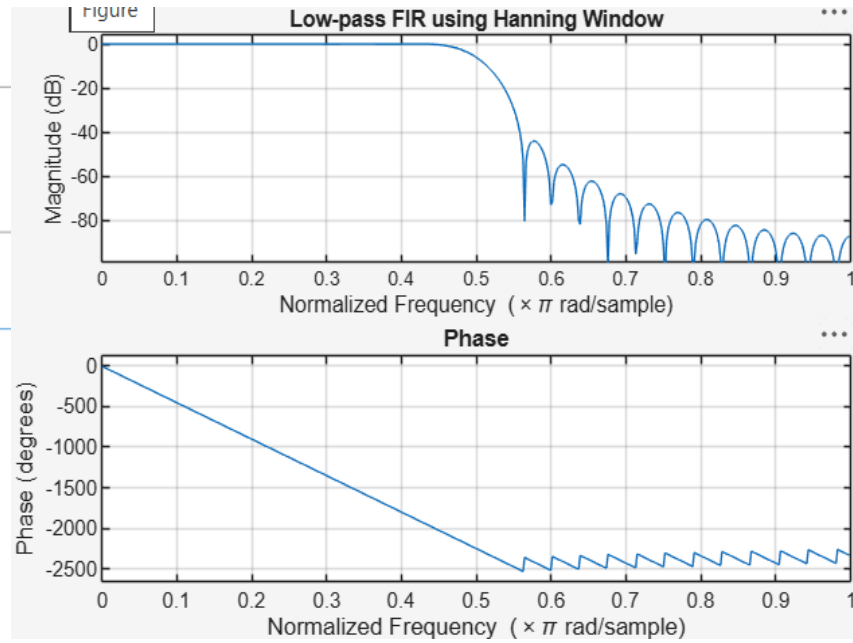
Exp 4.3

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % Passband cutoff (Hz)
fs = 4000; % Sampling frequency (Hz)
N = 50; % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
```

```
%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);
```

```
%% --- Plot Frequency Responses ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```



Exp 5.1

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

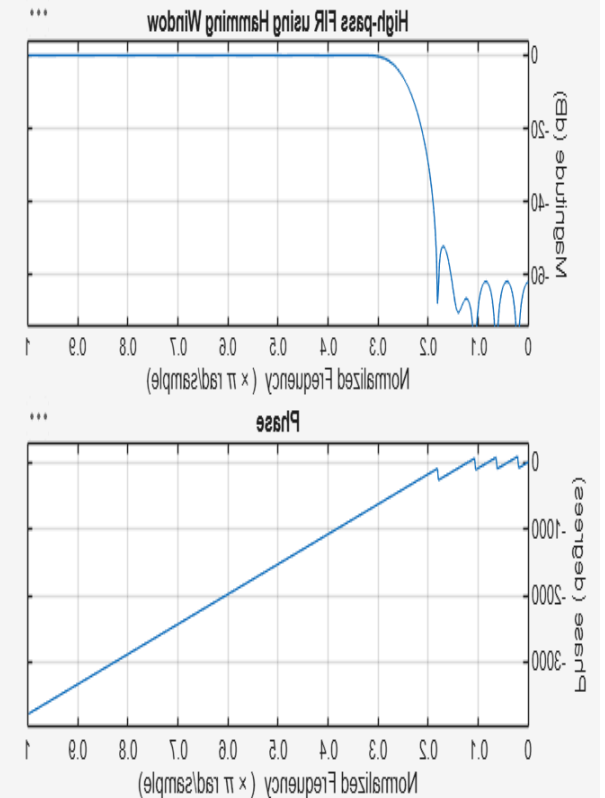
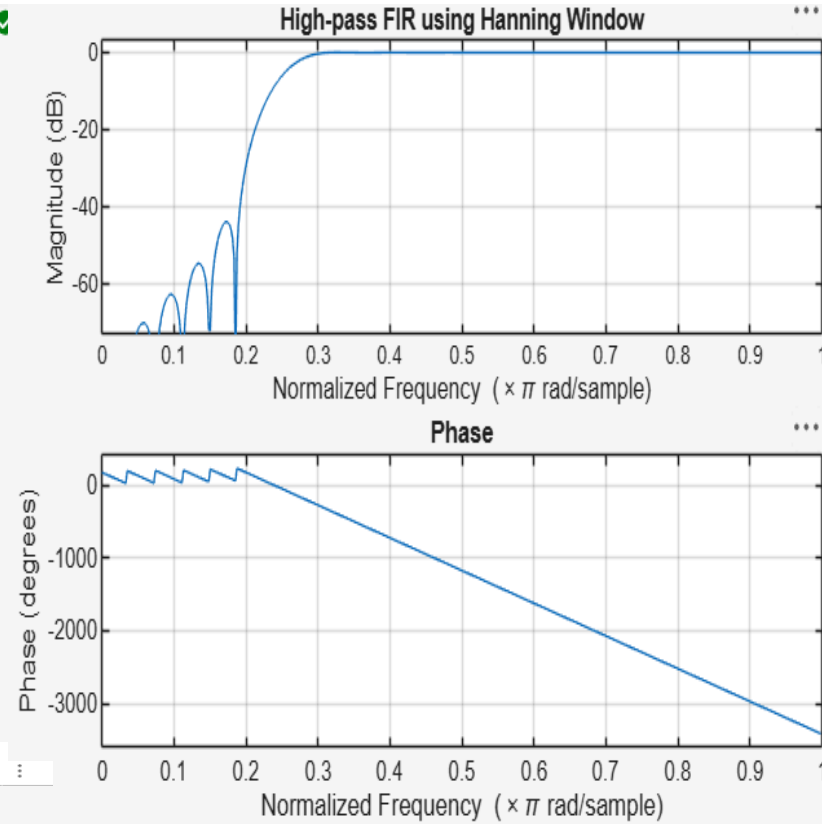
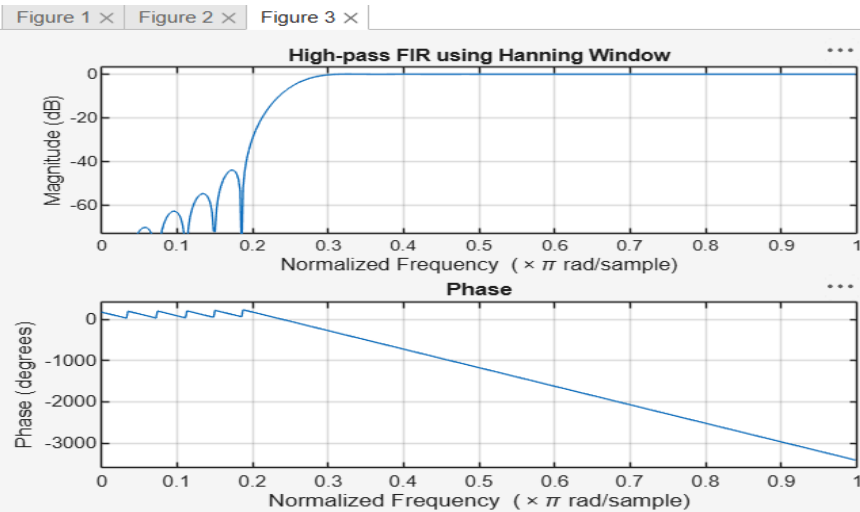
wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect');

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');

%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
```



EXP 5.2

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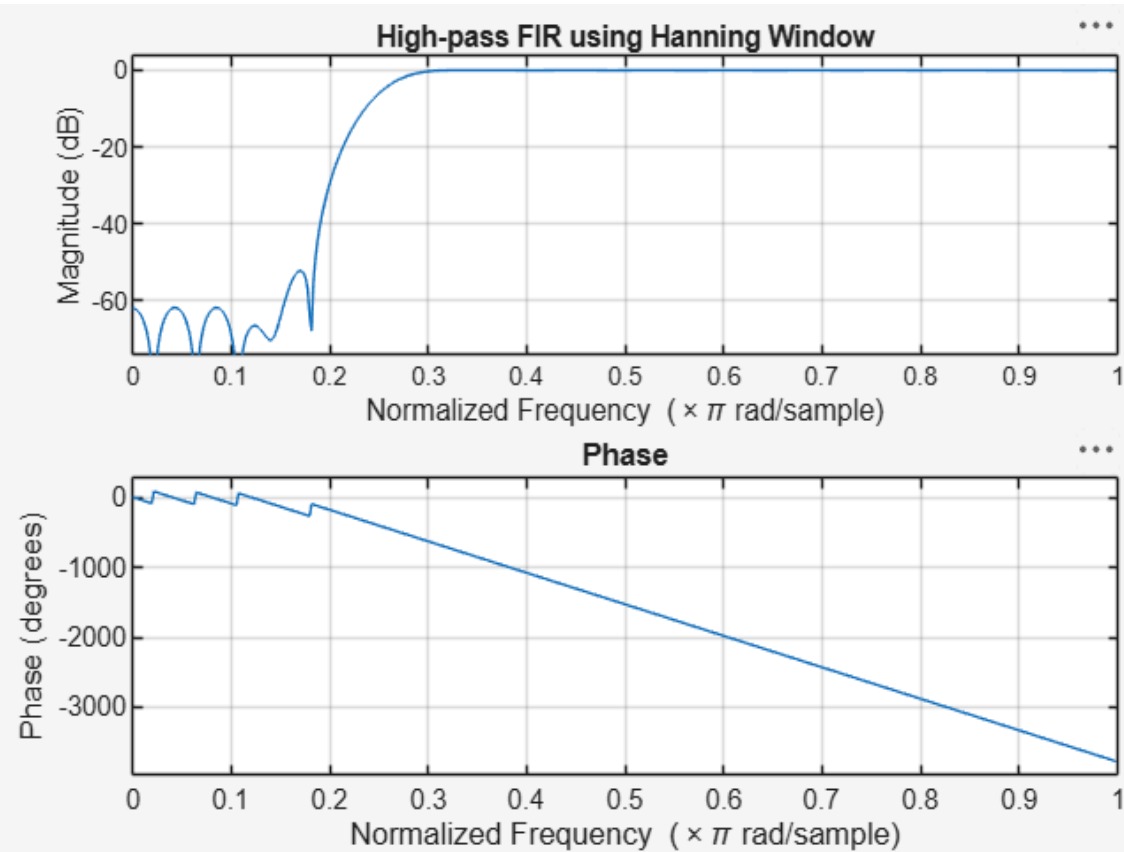
```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plots ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```



Exp 5.3

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0-1)
%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

